

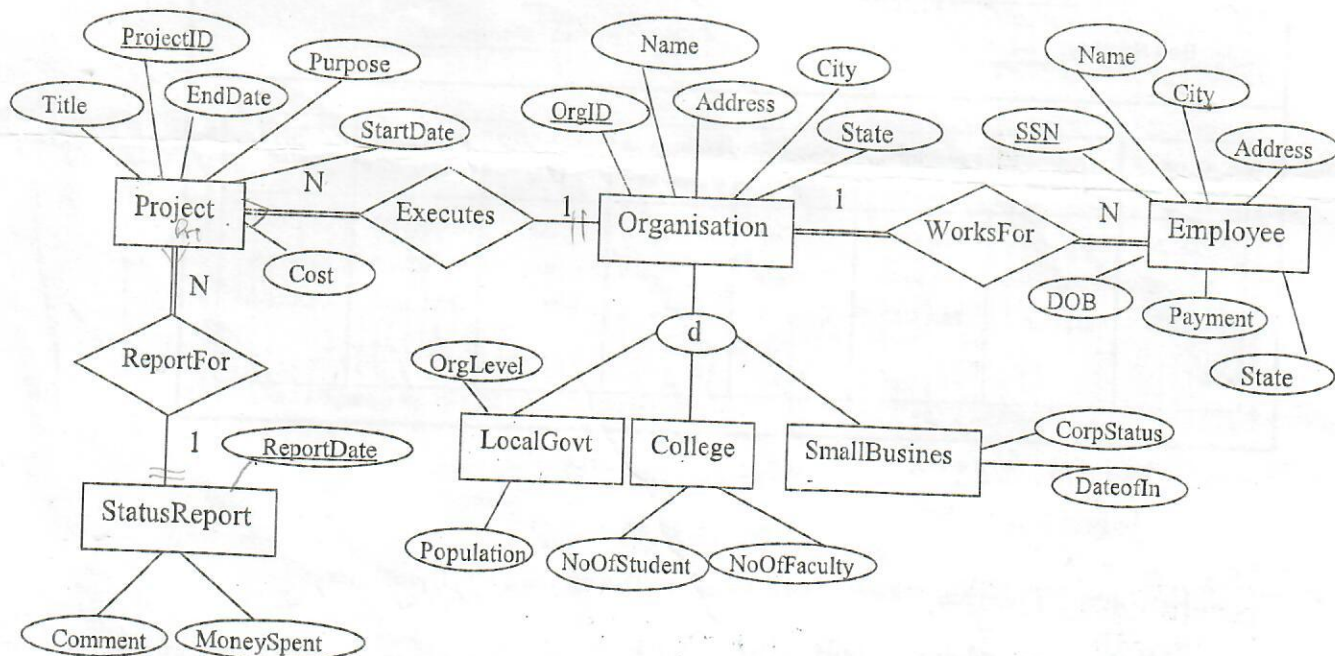


COVENANT UNIVERSITY

CANAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA.

TITLE OF EXAMINATION: B.Sc EXAMINATION
COLLEGE: SCIENCE AND TECHNOLOGY
SCHOOL: NATURAL AND APPLIED SCIENCES
DEPARTMENT: COMPUTER & INFORMATION SCIENCES
SESSION: 2020/2021
COURSE CODE: CSC 214
COURSE TITLE: HIGH PERFORMANCE AND DATABASE MANAGEMENT SYSTEM
INSTRUCTION: ANSWER ALL QUESTIONS
TIME: 1 HOUR

1. Convert the following ER diagram into a set of tables (logical Design).



2. The following tables form part of a database held in a relational DBMS:
- Hotel (hotelNo, hotelName, city)
 - Room (roomNo, hotelNo, type, price)
 - Booking (hotelNo, guestNo, dateFrom, dateTo, roomNo)
 - Guest (guestNo, guestName, guestAddress)

where Hotel contains hotel details and hotelNo is the primary key; Room contains room details for each hotel and (roomNo, hotelNo) forms the primary key; Booking contains details of bookings and

(hotelNo, guestNo, dateFrom) forms the primary key; Guest contains guest details and guestNo is the primary key.

Write SQL statements to

- List full details of all hotels.
- List full details of all hotels in London.
- List the names and addresses of all guests living in London, alphabetically ordered by name.
- List all double or family rooms with a price below £40.00 per night, in ascending order of price.
- List the bookings for which no dateTo has been specified.

3. Examine the Patient Medication Form for the Wellmeadows Hospital case study shown in the Figure below

- Identify the functional dependencies represented by the attributes shown in the form
- State any assumptions you make about the data and the attributes shown in this form.
- Describe and illustrate the process of normalizing the attributes to produce a set of well-designed 3NF relations. (Use PatientNo, DrugNumber as primary key).

Wellmeadows Hospital Patient Medication Form							
Patient Number: <u>P10034</u>							
Full Name: <u>Robert MacDonald</u>		Ward Number: <u>Ward 11</u>					
Bed Number: <u>84</u>		Ward Name: <u>Orthopaedic</u>					
Drug Number	Name	Description	Dosage	Method of Admin	Units per Day	Start Date	Finish Date
10223	Morphine	Pain Killer	10mg/ml	Oral	50	24/03/04	24/04/05
10334	Tetracycline	Antibiotic	0.5mg/ml	IV	10	24/03/04	17/04/04
10223	Morphine	Pain Killer	10mg/ml	Oral	10	25/04/05	02/05/06

Patient
Full Name
No

Bed/Ward/Ward
No No Name

DrugNo, Start Date
PatientNo, 11

DrugNo, WardNo
PatientNo,

1) PatientNo, DrugNo, WardNo, WardName, Full Name, Bed No, DrugName,
Desc, Dosage, Admin, Units, Start, Finish

WardNo



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DEPARTMENT: COMPUTER & INFORMATION SCIENCES

SESSION: 2020/2021

SEMESTER: ALPHA SEMESTER

COURSE CODE: CSC214

CREDIT UNIT: 3

COURSE TITLE: HIGH PERFORMANCE AND DATABASE MANAGEMENT SYSTEM

INSTRUCTION: ATTEMPT ANY FOUR QUESTIONS

TIME: 3 HOURS

QUESTION ONE

- (a) List 3 disadvantages of the File-Based System [3 marks]
- (b) Using the schema below, generate SQL statement for the following queries:
- Hotel (hotelNo, hotelName, City)*
Room (roomNo, hotelNo, type, price)
Booking (hotelNo, guestNo, dateFrom, dateTo, roomNo)
Guest (guestNo, guestName, guestAddress)
- (i) How many hotels are there?
(ii) List full details of all hotels in London
(iii) List number of rooms in each hotel [4.5 marks]
- (c) Use the following business rules to create the ERD for a small consulting company named ConsultCo. Show all entities, relationships, optionalities, connectivities and cardinalities. A client can contract many projects, but each project is contracted by one client. Some clients in the company client list have not (yet) contracted any project(s). A project may have many (consulting company) employees assigned to it, and each employee may be assigned to many projects. Some employees are not assigned to any project. One employee leads each project. An employee may lead more than one project at a time. One employee coordinates each project. An employee may coordinate more than one project at a time. Some employees never lead or coordinate projects. Many of the consulting company employees have multiple job skills that are applicable to the consulting work. Each job description- system analyst, database designer, accountant and so on- can fit many employees. Some job descriptions are not yet filled. Some employees do not have a specific job classification. [10 marks]

QUESTION TWO

- (a) Describe the following terms
- (i) Full functional dependency [2 marks]
(ii) Partial functional dependency [2 marks]
(iii) Transitive functional dependency [1.5 marks]

- (b) Describe and illustrate the process of normalizing the table below to Boyce-Codd Normal form, giving the following information [12 marks]

Candidate keys:

PNo, ADate (Primary key)

ADate, ATime, SurgeonNo

StaffNo, ADate, ATime

Patient_Appointment Table

StaffNo	DentistName	<u>PNo</u>	PName	<u>ADate</u>	ATime	SurgeonNo
S011	Tony Smith	P100	James White	12-Sep-08	10.00	S15
S011	Tony Smith	P105	Jill Bell	12-Sep-08	12.00	S15
S024	Hellen Chris	P108	Tom Fred	12-Sep-08	10.00	S10
S024	Hellen Chris	P108	Tom Fred	14-Sep-98	14.00	S10
S032	Rose Kelvin	P105	Jill Bell	14-Sep-98	16.00	S15
S032	Rose Kelvin	P110	John Stone	15-Sep-08	18.00	S13

Functional dependency:

PNo, ADate, $\rightarrow$ StaffNo, ATime, DentistName, SurgeonNo

PNo $\rightarrow$ PName

StaffNo $\rightarrow$ DentistName

StaffNo, ADate $\rightarrow$ SurgeonNo

ADate, ATime, SurgeonNo $\rightarrow$ PNo, PName, StaffNo, DentistName

StaffNo, ADate, ATime $\rightarrow$ PNo, PName

QUESTION THREE

- (a) UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g. flight, truck), and a deliveryRoute. create an Entity Relationship (E-R) diagram that captures this information about the UPS system. Be certain to indicate identifiers and cardinality constraints. [9 marks]
- (b) Given the conceptual E-R model below, write the business rules associated with it. [5.5 marks]

